Centralization through Consolidation:
Restructuring the Chinese Rare Earth Industry*

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This paper focuses on recent industrial policy changes in the Chinese rare earth industry, and attempts to show how the Chinese central government is shifting strategies for this important resource industry. Since the late 2000s, the Chinese central government has pursued consolidation of the highly fragmented rare earth industry into one dominated by large state-owned enterprise (SOE) groups under the control of the central government. This is far from liberalization and marketization as most media coverage and scholars have suggested. The Chinese central government is pursuing the centralization of the industry through nation-wide consolidation, which can make its intervention more effective and enhance the predominance of large SOEs under the name of efficiency.

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I. Introduction

By December 11, 2016, China had been a member of the World Trade Organization (WTO) for fifteen years. Since its accession to the WTO in 2001, debates have continued regarding whether China's greater integration with international economy has facilitated important changes in the Chinese economy, such as economic liberalization and marketization (Lardy 2002; Moore 2002; Zweig 2002; Panitchpakdi et al. 2002; Cass et al. 2003; Kennedy 2005; Naughton 2007). During the last decade, China has demonstrated signs that can be interpreted as such a transition in one of its most important strategic resource industries: rare earth industry.

China's rare earth policies have drawn keen international attention. Such attention is based on China's virtual monopoly status in the international rare earth market (Tse 2010; He 2014; UNCTAD 2014). Thanks to the Chinese government's aggressive export promotion since the mid 1980s, the Chinese rare earth industry acquired large production capacity and price competitiveness. Since the mid 2000s, China has supplied more than 90 percent of the world's rare earth consumption and between 2005 and 2008 supplied even more than 95 percent of global demands (Tse 2011; Tse 2012). China's rare earth policy changes, therefore, can have significant impact on international rare earth supplies and prices. For example, in 2011, when the

Chinese government halved its rare earth export quotas, international rare earth prices soared maximum seven times than the prices in 2010.

To make China's monopoly even more critical, rare earth elements (REEs) are vital ingredients in high-tech industries such as semiconductor, laptops, renewable energy and missile defense system. Supply problems in REEs can cause serious damages not only to high-tech industries but also to the entire national economies and security of advanced countries (UNCTAD 2014; Humphries 2015). Therefore, concerns have risen that China might use the high vulnerability of importing countries to the Chinese rare earth supply as a leverage in times of conflicts or crises (Hurst 2010; Morrison et al. 2012; Biedermann 2014). Such apprehensions were proven not unfounded. In September 2010, a Chinse fishing trawler collided with the Japanese Coast Guard patrol boats in disputed waters near the Senkaku/Diaoyu Islands and the captain of the Chinese trawler was arrested. To repatriate the captain, the Chinese government allegedly put an embargo on its rare earth exports to Japan, which left the Japanese government with no other choice but to comply.

Despite growing international criticism and anxiety, the Chinese government has continued imposing export restrictions on REEs since 1998. Such an aggressive government intervention seems to be a rational choice, considering the great strategic value of rare earth as well as soaring domestic demands and concerns over resource depletion that accompanied

China's rapid economic development. Interestingly, however, the Chinese government began to make a big shift in its rare earth industrial policies in the late 2000s, which, at least on the surface, can be interpreted as a move toward trade liberalization and market-oriented reform. In 2009, the Chinese government initiated industry-wide restructuring, which emphasized market rationalization and efficiency enhancement and dramatically reduced the number of rare earth production companies, and announced to abolish its decade-long export restrictions on REEs in 2015 (CREI 2015e; CREI 2016d).

This paper focuses on China's comprehensive rare earth industrial restructuring that has been pursued since 2009. Considering the strategic value of REEs and China's monopoly in the global market, the rare earth industry would be one of the most unlikely targets for industrial restructuring. Why then did the Chinese government pursue a dramatic policy shift? Can we translate China's rare earth industrial restructuring as a signal for its industrial marketization, even in industries with critical strategic values?

Despite the great security and economic importance of China's role in the global rare earth market, there are surprisingly limited amount of scholarly analyses on the Chinese rare earth policies. The majority of existing studies on China's rare earth policies tend to focus on the motives behind China's export restrictions on REEs, such as trade protectionism and national security measures, and rarely delve into domestic industrial

policies (Moran et al. 2008; Humphries 2010; Hurst 2010; Ruttinger et al. 2010; Kim 2011; Humphries 2013; Ting et al. 2013). There exist some studies that investigate the domestic sources of the Chinese rare earth policies (Jin et al. 2011; Hayes-Labruto et al. 2013; Wubbeke 2013; Biedermann 2014; He 2014), but they simply provide narratives on the individual policies and fail to analyze the logic of policy changes.[1] Although the Chinese rare earth industrial policies have significantly changed since 2009, only a few scholars have paid attention to the changes (Kim 2011; Malik 2013; Pothen et al. 2015) not to mention failing to analyze the mechanism behind the policy change.

Analyzing the background, contents, and progress of rare earth industrial policy changes during the last decade, this article attempts to analyze what China is attempting to achieve through dramatic industrial restructuring and streamlining in one of its most important resource industry, and gain insight on how China is transforming policies for strategic industries. In the following section, this paper shows how China's rare earth policies have evolved with increasing difficulties. Highlighting illegal mining and smuggling rampant in the Chinese rare earth industry, it demonstrates that the decade-long export restriction policies have been largely ineffective in alleviating the chronic problems. Section 3 analyzes the new industrial policies in the

1) Park and Jung (2015) is a rare exception. It explains why rare earth smuggling has persisted in China despite the central government's efforts to crack down the illegal practices, by analyzing the incentives of the local governments and producers.

light of this policy failure, and explains what made the Chinese central government eliminate export restrictions and instead pursue industrial consolidation.

This paper argues that the Chinese government's rare earth industrial restructuring during the last decade had originated from the continuous failure in export restrictions since the early 2000s. It has mainly aimed at establishing more effective central government control over the REEs industry by turning the industry into one dominated by large state-owned enterprise (SOE) groups that are better monitored and influenced by the central government. An interesting change is that the central government attempts to achieve such an industrial policy goal not simply by intensifying old-style direct intervention such as allocation of quotas but also by strengthening its regulatory power in taxation and environmental protection. This trend of industrial policy change is far from liberalization and deregulation of the Chinese rare earth industry as most media coverage and scholars have suggested. China seems to pursue "marketization" in a form that the central government sees efficient and desirable and where large SOE groups' market dominance is intensified.

Ⅱ. Policy Failure in the Chinese Rare Earth Industry[2)]

1. Rare Earth Policy Change: from Export Promotion to Export Restriction

When China began rare earth production in Inner Mongolia in 1959 (Humphries 2010; Tse 2010), the Chinese rare earth industry suffered from high production costs and low quality products and provided little threat to the competitors in the global market. With the beginning of the economic reform and opening in 1978, the Chinese government began to actively support rare earth production. This was because rare earth exports became a way to acquire foreign currencies and technologies and cement China's strategic ties with advanced economies (Parks 1981; He 2014). "Export promotion" became the primal goal of China's rare earth policy between 1985 and the early 2000s.

To promote REEs exports, the government provided diverse policy supports. Full tax rebates on REEs exports were introduced in 1985, and the National High Technology Research and Development Program was initiated in 1986 to facilitate REEs production (Kim 2011; Oh et al. 2011; He 2014). Thanks to the favorable government policies that eased small-scale,

2) For more details on the history of the Chinese rare earth policies and the smuggling problems, see Park and Jung (2015).

private rare earth producers' market access, by reducing their financial burden and helping them acquire necessary technologies, China's annual production of REEs in 1988 reached 29,640 tons, an amount tripled from a decade before (Tse 2010; Jin et al. 2011). By the mid 1990s, China's REEs production surpassed that of the U.S., making China the world largest REEs producer (Tse 2002; Hurst 2010). Other countries with REEs reserves, such as the U.S., Canada, and France, stopped mining because they could not match China in terms of price competitiveness. China soon gained a virtual monopoly in the global rare earth markets.

Around the turn of the century, however, the Chinese rare earth policies made a sharp turn from export promotion to "export restriction." Such a policy shift was mainly due to soaring domestic demands for rare earth products that accompanied China's rapid economic development. In 2005, China's domestic REEs consumption reached 52,000 tons, more than doubled from 20,000 tons of consumption in 2000. With such a dramatic increase in domestic demands, the ratio of China's domestic demands to domestic production has also jumped from 26 percent in 2000 to about 44 percent in 2005, and then again to 80 percent in 2012 (Morrison et al. 2012; UNCTAD 2014; Park and Jung 2015). By 2007, the Chinese domestic demands amounted to 73,000 tons, approximately equivalent to the total rare earth demands from the rest of the world, and in 2009 China's domestic consumption exceeded the

demands from the rest of the world. Since then, China has continued to consume the largest part of annual global REEs production.

With this dramatic increase in domestic REEs consumption, concerns surfaced that, if the current trend of production and exportation continued, China might face the depletion of its precious strategic resource and need to import REEs to meet its domestic demands in the future (State Council 2011; Tse 2010; Morrison et al. 2012). Thus the Chinese government began to implement a series of policies intended for export restriction since the early 2000s. Annual export quotas allocated to REEs producers by the Ministry of Commerce (MOC) since 1998 had been reduced over time. In 2005, the tax rebates on REEs exports were abolished (Mancheri 2015). In the following year, the Chinese government started to impose export duties on REEs products.

2. Failure of Export Restriction Policies

However, export restriction policies had been largely ineffective in terms of restraining REEs outflow. Ironically, this was mainly due to the success of production and export promotion policies during the earlier period, which had exponentially increased the number of small-scale rare earth mines in China and thus created a highly fragmented industrial

structure.

According to an interview by an official of the China National Nonferrous Metals Industry Corporation, as of 2009, there were as many as 20,000 small-scale, private REEs producers in China (China Daily 2009/09/03; Want China Times 2012/04/19; Park and Jung 2015). Faced with the new rare earth export restriction policies such as annual export quotas and export duties, these small-scale, private REEs producers did not reduce their production volume or close down mines. Instead, they resorted to illegal mining and smuggling. Illegal mining was particularly rampant in the dispersed deposits in South China where small-scale producers were working in opencast mines (Cheng et al. 2010; Jin et al. 2011; Kim 2011; Wubbeke 2013). Considering the already existing large production capacity and the wide gap between international and domestic REEs prices, it seems to be a rational choice for the local producers to continue production through illegal mining without official production quotas and sell their production abroad through smuggling.

Smuggling has been more rampant in southern China than in northern China (Wubbeke 2013; He 2014; Park and Jung 2015; Park 2016). This is closely related to the different geological distribution in rare earth mines and producers. The northern area has relatively concentrated industrial structure in which a large central state-owned enterprise in Inner Mongolia has been the biggest producer, while the southern region has a more fragmented structure in which the small and medium-size local

government-owned or private producers have been prevalent (Hurst 2010; Humphries 2013). Therefore, the southern regions have been more vulnerable to illegal practices such as smuggling or illegal mining (Park 2016).

It is not easy to estimate the amount of REEs smuggled out of China, but one can see the big picture by putting together different sources available in and outside of China. Statistics collected by foreign customs show that the volume of rare earth products imported from China between 2006, 2007 and 2008 were respectively 35 percent, 59 percent, and 36 percent higher than the annual volume officially exported from China under the export quota system (Park and Jung 2015). This suggests that between 2006 and 2008 around one third of Chinese official rare earth export quotas or approximately more than 20,000 tons of REEs per year were leaked out of China through illegal channels (Tse 2009; Hurst 2010; Tse 2010; State Council 2011; Tse 2011a; Morrison et al. 2012; State Council 2012; Wubbeke 2013; He 2014). Moreover, according to the estimation by Park and Jung (2015), despite the decrease in export quotas and the amount of official REEs exports since 2006, the amount of smuggling has not diminished. In other words, the volume of smuggling remained high, almost independent of the changes in the official export quota. This finding reveals how ineffective the export restriction policies have been in terms of reducing the amount of illegal REEs outflow.

Such a policy failure was not because the Chinese central

authorities have been indifferent to or condoned the illegal activities of the REEs producers. The Chinese central government had repeatedly expressed its will to crack down on illegal mining and smuggling since the mid 2000s (China Daily 2009/09/03; MIIT 2009; CREI 2010; New York Times 2010/12/29; State Council 2011; CREI 2012; Xinhua News Agency 2015/08/08), but its efforts did not paid off. An important reason behind the central government's continuous failure was the local governments' non-cooperation based on their economic protectionism (He 2014; Park and Jung 2015).

China's fiscal reform in 1994 was a key institutional change behind the intensified local economic protectionism (Whiting 2001; Naughton 2007). Through fiscal recentralization, the 1994 fiscal reform effectively increased the central government's share of local tax revenues, yet considerably reduced the local governments' revenue and made them rely more on the central government's revenue transfers than before. With no right to levy new taxes or change the rates on existing taxes, local governments have strong incentives to protect their existing revenue sources while defying central government policies that could diminish revenue sources (Economy 2007; Naughton 2007; Gang 2009; Ruttinger et al. 2010; Park and Jung 2015). In the rare earth-abundant inland regions with few other resources, the production and exportation of REEs have a huge share in the local economies. In addition, considering the relatively little production costs and enormous profits, it would be a lot more

beneficial for the local governments to acquiesce the illegal activities of the small-scale REEs producers in their jurisdictions or even help them continue with their illegal production and smuggling than to cooperate with the central government's initiatives to crack them down.

Su Bo, vice minister of Industry and Information Technology, had once complained that the local governments' protection had made the central government's crackdown on illegal operations and smuggling ineffective (CREI 2014a; Reuters 2014/01/09). Local governments have also protected rare earth producers against problems caused by pollution as well. Despite the severe environmental damages caused by REEs processing, local governments usually neglect to impose purification costs on REEs producers in order to provide them with economic advantages (MEP 2009; CREI 2014b; MEP 2014; CREI 2015b). Enticed with economic returns, local officials in some places even formed cooperative relationship with local rare earth companies. Some officials would purchase company shares while arresting villagers protesting pollution caused by REEs production and covering up companies' raids against the protestors (Lui et al. 2013; He 2014; Park and Jung 2015).

As Park and Jung (2015) had shown, given the highly fragmented and decentralized structure of the rare earth industry, local governments had strong incentives to condone illegal but lucrative REEs production and smuggling while circumventing or defying the central government's attempts to

control REEs production and outflow. When the center squeezed export quantity, small-scale local producers sought alternative ways to sell their overproduction by smuggling. Export restriction policies were proven ineffective, and this policy failure paved a way to a new policy shift: from export restriction to production control by reforming the entire industrial structure.

Ⅲ. Restructuring the Rare Earth Industry: Consolidation and Centralization

The Chinese central government pursued export restriction on rare earth for more than a decade without much success. Recognizing the failure, since the mid 2000s, the Chinese central government began to implement a new strategy, and switched its policy focus from export control to production control. The export restrictions on REEs, both export quotas imposed since 1998 and export duties imposed since 2006, were abolished in 2015. The new approach aimed at strengthening the central governments' sway over the rare earth industry by consolidating the fragmented and decentralized structure into one dominated by a small number of large state-owned enterprises (SOEs) under the centralized control and monitoring as well as

implementing a coordinated nation-wide industrial policies for the industry. The essence of this rare earth industrial restructuring can be summarized as "centralization" through "consolidation."

1. Consolidation and Centralization of the Rare Earth Industry

The consolidation of the fragmented rare earth industry was carried out by the merger of the small and medium-scale companies into large scale business groups. In 2010, two large business groups based on the regional distribution of REEs were established: the Northern and Southern Groups of Rare Earth Production.

The creation of the Northern Rare Earth Group was the result of the integration and reorganization of eight rare earth smelting and separation enterprises under the leadership of one SOE, the Baotou Steel Rare Earth High-Tech Co. in Inner Mongolia (CREI 2016a; CREI 2016c; CREI 2016g). The Southern Group of Rare Earth Production consisted of five SOEs in Jiangxi, Hunan, Fujian, Guangdong, and Guangxi, i.e. the China Minmetals Corporation, the Aluminum Corporation of China (Chinalco), the Guangdong Rare Earth Group, the Xiamen Tungsten, and the Ganzhou Rare Earth Group (MIIT 2009; Oh et al. 2011; He 2014; CREI 2015a). The China Southern Rare

Earth Group had 24 subsidiary companies with total registered capital of 1 billion yuan by the end of 2015, and formed a complete industrial chain covering exploration, mining, smelting & separation, processing, trade as well as research and development. As of 2016, the mining capacity of the Southern group reached 43,600 tons, with smelting & separation capacity of 59,700 tons (CREI 2016a). By the end of 2015, the once highly fragmented Chinese rare earth industry was restructured into six SOEs under two large business groups (CREI 2015e; CREI 2015f; CREI 2016a; CREI 2016g).

At least on the surface, the consolidation process seems effective. The number of companies involved in rare earth mining and operations has been reduced from 169 in 2010 to 20 as of 2015. As of the end of 2016, the six conglomerates had integrated 22 of 23 rare earth mines, incorporated 66 rare earth mining licenses, and 77 out of 99 smelting & separation enterprises (CREI 2016c). According to the MIIT's list of rare earth enterprises with backward capacity and overcapacity, announced in June 2014, 28 enterprises with total 103,710 tons of backward capacity were to be eliminated (Liu et al. 2015). In 2015, 40 rare earth smelting enterprises nationwide were closed and stopped production, which resolved excess capacity of 100,000 tons, one quarter of China's total production capacity, and reduced smelting and separation capacity from 400,000 tons of 2010 to 300,000 tons (CREI 2016g).

In addition, the consolidation measures have significantly

enhanced the central government's control over the rare earth industry. Before the consolidation, the implementation of export quotas and production quotas were mainly in the hands of the local governments. Even though the central government was in charge of annual quota allocation, without the local governments' cooperation with the implementation the quota system hardly functioned well. (State Council 2011; State Council 2012; CREI 2015e). However, under the current consolidated industrial structure, the central government directly allocates annual production quotas to the six large SOEs (CREI 2015b). There is little room for the local governments to maneuver the policy implementation process since the channels of production and exportation have been unified under the central government and the six SOEs (Park 2016).

As of the end of 2016, the consolidation seems to be in its final stages. The Chinese central government asserts that the industrial consolidation has significantly increased upstream industrial concentration and facilitated the centralized industrial chain covering the exploration, mining, separation, smelting, processing and trade of rare earth. In addition, the central government argues that, thanks to the consolidation, rare earth resource allocation became more reasonable and the market order has improved significantly (NDRC 2012; CREI 2015e; CREI 2016a; CREI 2016b). The actual effects of the consolidation measures should be a subject for further observation and analysis.

Despite the central governments' heavy emphasis on efficiency of production and market rationalization as the main achievements, the ultimate goal of industrial restructuring in REEs industry was not mere production efficiency or market rationalization in the rare earth industry. The essence of the industrial consolidation was to incorporate the highly fragmented industry under the protection of the local governments into an industry dominated by large SOEs under more direct control of the central government. The consolidation policy seems to be a strategic tool for creating a new market order in the industry according to the central government's preference, as it did in other strategic industries such as automobile or telecommunication (Pearson 2005; Pearson 2007; Hsueh 2011). China Rare Earth Information (CREI), published by the China Rare Earth Information Center, which is a Chinese government-owned research institute, frequently mentions "the rational utilization of rare earth elements and the regulation of market order" as the goals of REEs industrial consolidation. These goals suggest that the Chinese central government has identified irrational utilization of REEs or overproduction as a key problem in China's rare earth industry, and attempts to solve the problem by establishing "the orderly market competition" through its role as the regulator in the rare earth market (Lin et al. 2013).

In other words, new rare earth policies aimed at achieving centralization through consolidation. The consolidation of the

fragmented rare earth industry into large SOE dominant structure did not simply intended for better production control per se but ultimately endeavored to "rationalize" the entire industry, by building a coherent industrial structure that is under closer monitoring and control of the central government and facilitate the effective implementation of the central government's industrial policies.

2. Policy Tools for Industrial Restructuring

To successfully achieve industrial consolidation and ensure its control over the rare earth industry, the central government utilized diverse policy tools, such as production quotas exclusively allocated to the six conglomerates, new resource tax that substituted the abolished export duties, intensified environmental protection measures, and a crackdown on illegal mining and production.

First of all, the allocation of production quotas has been a key direct measure utilized by the central government to streamline the industry. Production quotas for the mining and operations of REEs were given almost exclusively to six large SOEs, making them control 95 percent of the allocated quotas in the entire rare earth industry and driving small-scale enterprises with no production quotas out of the market (NDRC 2011; NDRC 2012; CREI 2015e; CREI 2016b). For the first

batch in 2016, the MIIT assigned 52,500 tons for rare earth mineral production quotas and 45,000 tons for smelting and separation production quotas (CREI 2016d). Among them, production quotas for the six rare earth conglomerates were 52,400 tons of mineral products and 44,805 tons of smelting/separation products, which are almost the entire quotas.

According to the MIIT notice, the production quotas shall be allocated to enterprises that meet the national rare earth industrial policies and development plan requirements (MIIT 2009). The companies in any of the following situations cannot get the production quotas; mineral enterprises that have no mining license or work safety license or have been required to cease production by relevant departments; smelting/separation enterprises that still apply prohibited backward production techniques or have smelting/separation capacity less than 2,000 tons per year; the enterprises that cannot meet "rare earth industrial pollutants emission standards"; and new enterprises that were not allocated quotas in the past (CREI 2016d, 4). The notice also stresses that enterprises that are found conducting violations, overproducing, and causing environment pollution and potential safety hazards yet have not completed rectification will face reduced or cancelled quotas. The notice also proposed to strengthen operation monitoring. Six rare earth groups should set up production accounts and report the monthly production and operation situation before the 5th of each month (CREI 2016j).

Other important yet relatively indirect policy tools are

taxation and environmental regulations. Their main roles regarding industrial consolidation was increasing production costs. In the process of consolidating the fragmented rare earth industry, carried out by the merger and acquisition of the numerous small and medium-scale companies, a measure utilized to facilitate the merger and acquisition was increasing production costs for REEs. The central government regarded that low production costs have made it easier for the small-scale producers to engage in illegal mining of REEs. The "Review on 2014 China Rare Earth Policies" issued by China Rare Earth Information Center sees resolving the long-standing problem of the low costs of illegal production as the way "to make rational utilization for rare earth resources and to regulate the market order" (CREI 2015c).

In the past, the small-scale producers did not have to pay any value-added taxes, resource taxes, and waste disposal costs, despite the fact that the REEs processing was a main cause of severe pollution and residents' health problems in the regions producing REEs. In May 2015, the State Council first imposed volume-based resource tax on REEs production, and then raised the tax burden again by deciding to levy price-based resource tax in April 2016 (CREI 2016e; CREI 2016k). Price-based resource tax uses a higher tax rate than the volume-based one to begin with and increases according to the fluctuation of price, therefore has the effect of tax increase. Imposing resource tax on REEs production increased financial

burdens on small-scale REEs producers that will eventually make them wither or merged into large SOEs (State Council 2011; Park 2016).

Environmental regulations have served as a useful supplementary measure to fortify the central government's control over the rare earth industry as well. A package of environmental protection measures has been issued during the last couple of years. The MIIT released "the Clean Production Technology Plan for Rare Earth Industry" in February 2014, requiring REEs producers to utilize clean technologies for wastewater purification to reduce the emission of CO2 and ammonia-nitrogen (MIIT 2014a; CREI 2015c). "The 2014-2015 Energy Conservation and Emissions Reduction and Low-carbon Development Action Plan," released by the General Office of the State Council in May 2014, stated reinforced emission and carbon reduction indexes for REEs enterprises. With these new environmental measures introduced, the MIIT and the MOC allocate the production quotas only to those who meet environmental requirements and thus greatly strengthen the central government's monitoring and control power over the rare earth enterprises.

In addition to the above mentioned policy measures, the Chinese central government has increased its direct intervention to eradicate illegal mining and smuggling, the chronic problems in the Chinese rare earth industry. According to government insiders, if the government cracks down on illegal mines, there

will be a 50% fall in the illegal mining output and about 40% decrease in the supply of rare earth minerals (CREI 2016f).

A multi-ministerial action plan has been orchestrated to crackdown on illegal production. The MIIT issued the "Notice on Specification and Administration of Rare Earth Resources Recycling Projects" in May 2014 (MIIT 2014b; CREI 2016d). According to the Notice, approval of rare earth smelting/separation projects pursued under the name of rare earth recycling should be standardized in order to prevent the local government's approval of illegal rare earth production. As a part of such a new policy, all local authorities are required to verify the source of rare earth wastes (CREI 2016c, 3-5). In addition, the MIIT, the Ministry of Public Security, the Ministry of Land and Resources, the Ministry of Environmental Protection, the General Administration of Customs, and the State Council jointly issued "the Special Action Plan on the Crackdown of Illegal Rare Earth Behaviors" in September 2014, and enforced stricter monitoring over rare earth production (CREI 2015c; CREI 2016a). According to the plan, a "special action" was implemented in three stages: a self-checking stage (from October 10 to November 25, 2014), a rectification stage (from November 26 to January 31, 2015), and an acceptance and summary stage (February 1 2015 to March 31, 2015). This action plan also imposed mineral resource tax, compensation fees and drainage charges as well as environmental restrictions on REEs producers to resolve the long-standing problem of illegal

mining by small-scale producers.

In 2015, MIIT together with other six related ministries conducted five times of special actions against illegal rare earth behaviors. Total of 67 cases were investigated and more than 12,400 tons of illegal rare earth mineral products were seized (CREI 2016f; CREI 2016g). During the 12th Five-Year Plan period, 14 illegal mines and 28 companies were closed. According to the MIIT statistics, more than 36,000 tons of illegal rare earth goods were seized, and 230 million yuan fines were imposed (CREI 2016h). It seems that such a strong government crackdown on illegal behaviors will continue. In August 2016, according to Zhou Changyi, head of the MIIT department of raw materials, Chinese ministries vowed to enhance cooperation and take zero-tolerance approach toward illegal mining of rare earth products during the 13th Five-Year Plan period from 2016 to 2020. A tracking system will be implemented to trace the sources of illegal products, he said (CREI 2016h; CREI 2016j).

Ⅳ. Conclusion: Implications for China's Industrial Policies

This paper focuses on the Chinese rare earth industrial policies during the last decade, and attempts to show how and why the Chinese central government has shifted its development strategy for this important resource industry with grave strategic importance and pursued dramatic industrial restructuring. During the 2000s, the Chinese central government had focused on imposing export restrictions such as export quotas and export duties, to control the entire rare earth industry and to slow down rare earth outflow and depletion. Such export restriction policies became a target of international criticism that China was pursuing resource weaponization and had violated international trade norms. In August 2014, the WTO ruled that the Chinese REEs export restriction policies had violated WTO obligations.

Considering this context, it is interesting that China had initiated a policy turn for rare earth industry that seemed to actively accommodate the international criticism against excessive government intervention. China has implemented a comprehensive nation-wide industrial restructuring focusing on production efficiency and market rationalization since the late 2000s, and finally abolished export quotas and duties on REEs in 2015. On the surface, China's industrial policy turn could be interpreted that it began to allow more marketization and trade

liberalization in the rare earth industry and to abide by WTO obligations. This is the dominant view that existing media coverage and scholarly analyses on the Chinese rare earth policies have suggested.

However, this paper provides an argument that contradicts such an interpretation. The Chinese central government has tried to increase its control over the rare earth industry by consolidating the industry into a structure dominated by large-scale SOE groups that are under tighter and more effective monitoring and control by the central government. Interestingly, the central government attempts to achieve such an industrial policy goal not simply through direct intervention such as setting production quotas and cracking down illegal activities but also through indirect regulation such as imposing resource tax and strengthening its regulatory functions in a new issue area, i.e. environmental protection. During this trend, the abolition of export restrictions was simply a move away from the old, direct intervention strategy that was proven inefficient.

It is intriguing and requires further analysis that such a shift toward consolidation and centralization have also been found in other strategic industries, which have suffered overproduction problems similar to the rare earth industry. Since 2006, the Chinese government has raised concerns about overproduction in key strategic industries such as steel, automobile, cement, flat glass, coal, and chemical industries (Bajona et al. 2009; Steel Industry Coalition 2016). In each industry, the government has

issued a series of policies to address the overproduction problem from the production side, a policy turn demonstrated in the rare earth industry. Such policies include putting limits on new production, eliminating outdated production facilities, and encouraging consolidation into large SOEs.

It is premature to determine whether the Chinese government's industry-wide attempts have been successful in eliminating the chronic overproduction problem in these industries. However, the policy changes in the rare earth industry seems to suggest that China's industrial policies are evolving toward establishing more centralized control through industrial consolidation based on large SOEs. The goal behind such policy switch is far from industrial liberalization and deregulation. The Chinese central government emphasizes "market" and efficiency, but it means the introduction of market order and structure that the central government sees desirable, that make its intervention more effective, and where the monopoly of large SOEs is predominant under the name of efficiency.

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통합과 중앙집중화: 중국 희토류 산업정책의 전환

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이 논문은 중국이 독점적 지위를 누리는 핵심 전략자원인 희토류 산업에서 지난 십여 년 동안 일어난 정책변화를 분석함으로써, WTO시대 중국의 전략산업 정책전환을 조명하고자 한다. 2000년대 후반 이후, 중국 중앙정부는 기존의 수출통제 중심의 희토류 산업정책에서 크게 선회, 산업구조의 합리화와 수출통제 폐지를 추진해왔다. 이 논문은 이러한 정책변화가 기실 자유화나 탈규제와는 거리가 먼 것으로, 파편적인 희토류산업을 통합과 중앙집중화를 통해 대규모 국유기업 중심으로 재편하고 산업전반을 보다 효율적인 중앙정부 통제 하에 두는 것을 목적으로 함을 보여준다. 이러한 희토류 산업정책은 중국이 전략적으로 중요한 국가 자원산업에 있어 중앙정부의 감독과 통제권을 강화하고 대형 국유기업의 독점체제를 만드는 형태로 정책을 전환하고 있음을 시사한다.

주제어: 중국, 희토류, 통합, 중앙집중화, 산업정책